

**LONG-DISTANCE MIGRANTS**

In Alaska and northern Canada, vast herds of reindeer migrate between their summer grounds on open tundra and their winter grounds in coniferous forest. Some travel over 620 miles (1,000 km) each way, swimming across rivers and sea inlets en route. Pregnant females lead the herd during the spring leg of the journey.

**WINTER COAT****SUMMER COAT****SEASONAL COATS**

When the Arctic fox changes color, the consistency of its coat changes as well. Its white winter coat has long guard hairs and thick underfur, making for superb insulation. Its brown summer coat is shorter, with thinner underfur, to help prevent overheating.

Summer migrants

Near the poles, 24-hour daylight in summer creates the ideal conditions for rapid plant growth. This short-lived but profuse supply of food has a dramatic effect on tundra life, attracting vast numbers of migrants. Geese come to crop the plants with their beaks, waders arrive to feed on worms and insects that live in swampy ground, while terns find food both on the tundra and in water close to the shore. This annual influx of visitors is mirrored in the ocean. Many of the world's baleen whales head toward polar waters during summer to make use of the annual upsurge in planktonic life. However, unlike migratory birds, these huge mammals do not breed at high latitudes.

Instead, they put on weight and then return to warmer waters to give birth. During the breeding season, they often do not feed at all.

Winter food

Although the food supply in polar seas slowly falls in the fall, there is still a reasonable amount for animals to eat. On land, life is not so easy. The growth of tundra plants comes to a complete halt and, to make matters more difficult, the plants themselves are often covered by deep snow. For herbivores, this lack of accessibility is a major problem at a critical time of year. In the Arctic tundra, plant-eating animals reach plants in one of 2 different ways. Reindeer (caribou) and musk oxen use their hooves to clear away the snow to reveal the lichens and dwarf willows underneath. Lemmings turn the snow to their advantage by burrowing in it. The snow protects them from predators and from the weather outside: no matter how cold or windy

it is on the surface, the lemmings enjoy a benign micro-climate that allows them to feed all year. In Antarctica, there are very few terrestrial plants, which means that almost no animals stay active in winter on food gathered from land. With so much ice, even food from the ocean can be difficult to reach. Male emperor penguins, guarding their eggs, do not even attempt to find it: huddling on the ice through the long night of winter, they go without food until spring.

**RUNWAYS UNDER THE SNOW**

Lemmings need cover to survive. In winter, they develop enlarged front claws that enable them to tunnel in snow. The tunnels keep them secure, although they must still contend with sparse winter food supplies. When the snow melts, they move underground.

TUNDRA INSECTS

Unlike Antarctica, the Arctic tundra teems with insect life, including aphids, bumble bees, damselflies, and unimaginable numbers of mosquitoes. The mosquitoes spend their larval and pupal stages in tundra pools, emerging as adults in early summer. For warm-blooded animals, tundra mosquitoes are a serious problem. The females need blood before they can breed, and they are relentless in their attempts to get it. When the mosquito season is at its height, some animals head for high ground, but most have no alternative but to sit it out. Mosquitoes do bring some benefits: the larvae and pupae are a useful source of protein for waterfowl and waders.

CLOUD OF MOSQUITOES

Adult tundra mosquitoes live for only a few weeks once they have emerged from their aquatic nurseries—just long enough to feed, mate, and lay eggs, before fall frosts glaze their breeding pools with ice.

